

📖 ?What is the correct order of operations of the OpenGL graphics pipeline

projection transformation 1

modelview transformation 2

viewport transformation 3

divide by w 4

.Write down the Modelview matrix after executing the following OpenGL code segment

```
glMatrixMode(GL_MODELVIEW)
glLoadIdentity()
glRotatef(90.0, 0, 0, 1.0)
glScalef(2.0, -1.0, -1.0)
```

A)	B)	C)	D)
$\begin{bmatrix} 0 & -1 & 0 & 0 \\ -2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & -2 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

A

B

C

D

Select the OpenGL commands that are used to define the projection matrix (Multiple Answers)



gluPerspective

gluLookAt

glOrtho

glProjection

glFrustum

.Write down the Modelview matrix after executing the following OpenGL code segment

```

;glMatrixMode(GL_MODELVIEW)
;()glLoadIdentity
;glRotatef(90.0, 0, 0, 1.0)
;()glPushMatrix
;glScalef (-1.0,0.5, -1.0)
;glTranslatef(-1.0f, -2.0f, -3.0f)
;glRotatef(-90.0, 0, 0, 1.0)
;()glPopMatrix
;glRotatef(90.0, 0, 0, 1.0)
;glScalef (-1.0,-1.0, 1.0)

```

A)	B)	C)	D)
$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

A

B

C

D

Let us scale a square with one unit side length using `glScale(2.0f, 0.0f, 0.0f)`. The output after  :scaling is

.Rectangle with two units' width and one unit height

.Rectangle with two units' height and one unit width

.The square will disappear

.Square with two units side length

.Square with one unit side length

$$m = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

 The OpenGL represents the following matrix as

`;GLfloat m[] = { 1.0f, 2.0f, 3.0f, 4.0f, 5.0f, 6.0f, 7.0f, 8.0f, 1.0f, 2.0f, 3.0f, 4.0f }`

`;GLfloat m[] = { 1.0f, 5.0f, 1.0f, 2.0f, 6.0f, 2.0f, 3.0f, 7.0f, 3.0f, 4.0f, 8.0f, 4.0f }`

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14

Generally, what primitive polygons are used for creating a mesh to represent a complex object? matrix (Multiple Answers)

Cube

Sphere

Lines

Circle

Triangles

Points

15

?What kind of argument is expected for glColor3fv command

A GLuint *pointer

Any void variable

.An array typecasted with (GLfloat *) or in the type indicated by the preceding letter

A virtual method function pointer

Q40 : Rotating the 3D point (1,1,0) by 90 around z-axis gives what

(0,1,0)

(-1,1,0)

(1,0,0)

(-1,-1,0)

(1,1,0)

(1,-1,0)

diffusion reduces the memory and computational cost by applying the diffusion

Q40 process over a lower dimensional space instead of using the actual pixel

CLIP

Latent

Scheduler

Stable



?What is the correct order of operations of the OpenGL graphics pipeline

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projection transformation 1

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